

Day 1: Power BI Fundamentals

📌 Definition to Remember

Power BI is Microsoft's unified enterprise business intelligence and data visualization platform that connects disparate data sources, transforms messy data into relational models, and produces interactive analytical reports and cloud dashboards.

1. Power BI Ecosystem & Architecture

- **Power BI Desktop:** Free Windows authoring tool used for data connection, Power Query transformation, DAX modeling, and report creation.
- **Power BI Service (SaaS):** Cloud-hosted collaboration platform (app.powerbi.com) used for publishing reports, building real-time dashboards, sharing workspaces, and scheduling automated refreshes.
- **Power BI Mobile:** iOS and Android apps for on-the-go dashboard consumption.
- **Power BI Report Server:** On-premises report server for organizations requiring internal hosting behind firewalls.

2. Core Concepts & Terminology

1. **Semantic Model (formerly Dataset):** The heart of Power BI containing data tables, relationships, data types, and DAX measures.
2. **Report:** Multi-page interactive analytical document built on a single semantic model.
3. **Dashboard:** Single-page executive summary board in Power BI Service composed of pinned visual tiles from one or multiple reports.
4. **Dataflow:** Reusable ETL data preparation pipelines built in Power BI Service using Power Query Online.
5. **Workspaces:** Collaborative cloud folders (My Workspace vs Shared Workspaces) where teams manage reports, datasets, and apps.

3. End-to-End Power BI Workflow

```
[Data Sources] (Excel/SQL/Web)
|
▼ (Extract & Clean)
[Power Query ETL]
|
▼ (Model & DAX)
[Data Model & Relationships]
|
▼ (Design & Visualize)
[Power BI Desktop Report]
|
▼ (Publish to Cloud)
[Power BI Service (App/Workspace)]
|
▼ (Consume)
[End Users / Mobile / Embedded]
```

★ Must-Master Skills & Interview Points

1. Differentiate clearly between **Report** (multi-page, single dataset, interactive slicing) and **Dashboard** (single-page, cross-report tiles, cloud only).
2. Explain the role of **Semantic Models** as the single source of truth.
3. Understand when to use **Power BI Desktop** (authoring) vs **Power BI Service** (collaboration, RLS enforcement, automated refresh).

⚡ Quick Recall

Connect Data → Clean in Power Query → Model Relationships → Calculate with DAX → Design Visuals → Publish to Service